

SECTION – A

There are **FOUR** questions in this section. **Question No. 1 (one) is Compulsory.** In addition to answering Question 1, answer any **2** other questions.

All the questions, except for Question 1, are of equal value. Symbols have their usual meanings

1. (a) Describe with necessary examples:

(15)

(i) One-hot encoding

(ii) Confusion matrix

(iii) Bias-variance tradeoff

(b) For the univariate data below, based on current parameter settings, the $p_{ij} = P(C=i|x_j)$ values have already been calculated in the **E-step** for **mixture of 2 Gaussians**. Produce the model parameter values in the **M-step**.

(15)

Data Id	Feature	$P(C=1 x_i)$
x1	49	0.25
x2	53	0.50
x3	45	0.25
x4	45	0.20
x5	51	0.15

(c) You are given a dataset composed of voice recordings containing 22 biomedical features from 31 people, 23 with Parkinson's Disease (PD). For each individual, voice recordings were taken approximately 6 times. However, during recording, sometimes some biomedical features were missed. Features from each recording constitute a row in the data file. A "name" column identifies the subject from whom the data was collected. A "status" column is set to 0 for healthy and 1 for PD. All the remaining columns contain real numbers (and some missing values). In total there are 195 rows and 24 columns in the dataset. 147 of the rows are marked 1 in the status column. The main aim of the data is to discriminate healthy people from those with PD.

(15)

(i) What pre-processing would you apply to this data? Elaborate.

(ii) Will you use deep learning or conventional machine learning algorithms? Explain your answer.

(iii) Will you use a separate validation set for hyper parameter tuning or use cross-validation? Argue why.

2. (a) Deduce the **normal equation** for multivariable linear regression. What is its limitation?

(10)

(b) What is ensemble learning? What are its advantages? Explain boosting with necessary examples.

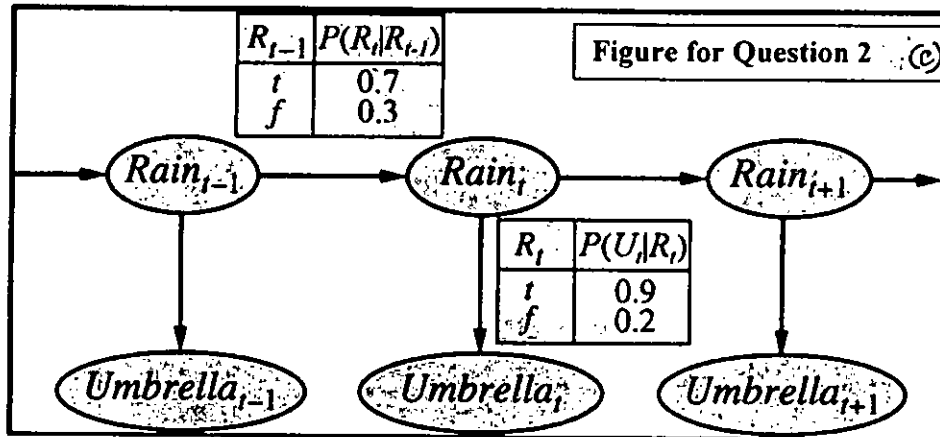
(10)

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Contd ... Q. No. 2

- (c) In the umbrella world described by the Bayesian network in the Figure for Question 2 (c), rain is the hidden variable, which needs to be inferred from the observation of umbrella. Calculate the smoothed estimate of rain probability on day 1, given umbrella observation on both day 1 and day 2. Assume a 50% chance of rain on day 0.

(10)



Bayesian network structure and conditional distributions describing the umbrella world. The transition model is $P(Rain_t|Rain_{t-1})$ and the sensor model is $P(Umbrella_t|Rain_t)$.

3. (a) Describe the process of parameter estimation in Hidden Markov model. (20)
- (b) Suppose you have N numbers $x_1, x_2, \dots, x_N$ drawn from a Gaussian distribution. Recall that the Gaussian density function on a single variable is: (10)

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This linear-Gaussian model has two parameters: mean (μ) and standard deviation (σ). Derive the maximum-likelihood estimates of these two parameters.

4. (a) For a K -class classification problem, there are K neurons in the output layer of a deep network, with outputs of $f_1, f_2, \dots, f_K$ respectively. Then a softmax layer is added to normalize these into class probabilities $p_1, p_2, \dots, p_K$ respectively. Cross entropy loss is used to train the model. For the i^{th} data point in the training set (x_i, y_i) , Where x_i represents the feature vector and y_i its one-hot encoded class label, state the expression for loss (L_i), Then deduce the expression for $\frac{\partial L_i}{\partial f_k}$. (10)
- (b) What is a non-linear activation function? Justify its use in a deep network. Give three examples of such functions, with their definition and graphical representation. (10)
- (c) Explain Adagrade optimizer with necessary equations. What are its strengths and weaknesses? Explain with necessary equations how RMSProp improves on it. (10)

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SECTION – B

There are **FOUR** questions in this section. **Question No. 5 (five) is Compulsory.** Answer

Question 5 and any other **TWO** questions.

Question 5 must be answered

5. (a) Explain the four assumptions in the context of AnyNet design space which helps to explore the network architecture space effectively. (12)

- (b) Modern convolutional neural networks (CNN) evolved from one architecture to another to adapt to increasingly complex application scenario. Some network pairs are mentioned below. (12)

For each network pair, illustrate the modification done to get the second network from the first, and the reason behind the change.

First Network	Second Network
AlexNet	VGG
VGG	NiN
NiN	GoogLeNet

- (c) During deep neural network training, at some stage it might get difficult to understand whether an additional layer/block (increased depth) will provide additional performance boost or not. Devise a technique by which we can let the network decide whether to include that additional layer/block or not. (11)

6. (a) To train a recurrent neural network (RNN) we use BPTT (Backpropagation Through Time) algorithm. Let us consider a simple RNN without bias parameters, whose activation function in the hidden layer uses the identity mapping. The dynamics of this RNN are governed by the following equations where symbols have their usual meaning. (15)

$$h_t = W_{hx} x_t + W_{hh} h_{t-1}$$

$$o_t = W_{qh} h_t$$

Using chain rule, show that a large power of the matrix W_{hh} is responsible for vanishing/exploding gradient problem.

- (b) Draw the internal gating mechanism of an LSTM cell. Write the complete mathematical formulation of LSTM memory cell update. (10+10=20)

7. (a) Both word2vec and transformer architectures produce vector embedding of a word. What is the fundamental difference between these two embeddings? (10)

- (b) What is the difference in the context of latent space modeling between traditional autoencoder and variational autoencoder? How is this difference advantageous for a variational autoencoder? (10+5=15)

- (c) How do the gradients propagate through the sampling layer of a variational autoencoder while executing the standard backpropagation algorithm during training? (10)

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8. (a) Assume that a 2-headed self-attention layer within the first encoder unit of a transformer has the following learned parameter matrices [all the notations have their usual meaning].

(25)

$$\begin{aligned} W_0^Q &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, W_0^K = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}, W_0^V = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \\ W_1^Q &= \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}, W_1^K = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}, W_1^V = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ W^O &= \begin{bmatrix} 1 & 3 & 2 & 5 \\ 4 & 5 & 6 & 3 \\ 1 & 3 & 2 & 5 \\ 4 & 5 & 6 & 3 \\ 1 & 3 & 2 & 5 \\ 4 & 5 & 6 & 3 \end{bmatrix} \end{aligned}$$

The key-query dot-product should be normalized by dividing by 8 before passing through the Softmax layer. For the following word embedding and positional encoding matrices, calculate the output encoding of the self-attention layer. Show all the intermediate calculations and results.

$$X = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 4 & 5 & 6 & 3 \end{bmatrix}, P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

- (b) Why do we use multi-head attention in the transformer architecture?

(10)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Derive the parameter (theta) update equation, $\theta_j = \theta_j + \alpha ((y^i - h(x^i))x_j$ for stochastic gradient descent algorithm of binary logistic regression classification problem. Here all the symbols convey usual meaning. (15)
- (b) The linear regression has a closed form solution. Derive the closed form equation to find optimal theta for a linear regression problem. (10)
- (c) How does Boosting help in the bias-variance trade-off? Explain with an example. (10)

2. (a) Write down the pseudo-code of the Q-learning algorithm for the reinforcement learning strategy. Precisely discuss the role and intuition of each component of the update (state and action) equation. (15+10)
- (b) You are learning a target function, which is a sinusoidal curve (unknown). You are given two hypotheses: $H_0 : h(x) = b$, $H_1 : h(x) = ax+b$, and two data points. Which hypothesis should you pick as a learning solution based on bias-variance trade-off, and why? (10)

3. (a) What is the motivation of the support vector machine (SVM) based learning over a decision boundary (e.g., perceptron) based learning? Derive the equation that you want to minimize for getting the solution of the SVM. (5+15)
- (b) Draw a high level block diagram showing the interaction of different components of a learning problem. (10)
- (c) What is Markov Decision Process (MDP)? (5)

4. (a) How can you categorize different clustering methods? (6)
- (b) Write down the pseudo-code of the k-means clustering method. How can you find out a suitable k for the above clustering method? (8+6)
- (c) What is Softmax regression? Explain a three-class classification problem with an example to formulate the necessary equations of Softmax regression based classification solution. (3+12)

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Write down the mathematical definition of ReLU. Draw its curve and that of its first derivative. Is ReLU differentiable at all points? If not, then explain how can it still be used in neural networks? (10)

- (b) Explain, with mathematical reasoning, why *Xavier Initialization* samples weights from $N\left(0, \frac{2}{n_{in} + n_{out}}\right)$, where n_{in} and n_{out} are respectively the number of input to and output from a node in a hidden layer respectively. (10)

- (c) In an FNN, a mini batch of data is shown in the table below. Here f_i means the i -th feature of the data. What will be the normalized first data row seen by the network, in case of (15)

(i) Batch normalization (Assume $\gamma=1, \beta=0, \epsilon=0$)

(ii) Layer normalization (Assume $\epsilon=0$)

	f_1	f_2	f_3	f_4	f_5
Data# 1	2	1	9	7	1
Data# 2	3	2	8	5	2
Data# 3	5	4	9	3	3
Data# 4	2	3	9	6	4

6. (a) With stride 2 and padding 1, calculate the transpose convolution of $\begin{bmatrix} 2 & 5 \\ 7 & 4 \end{bmatrix}$ using the

filter $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 3 \\ 1 & 0 & 1 \end{bmatrix}$. (10)

- (b) Explain Adam optimizer with necessary equations. (10)

- (c) Draw the schematic diagram of the memory cell internal state of LSTM. Explain the purpose of the different components, with necessary equations. Mention the dimensionality of the various outputs and weight parameters. (15)

7. (a) What is label smoothing? How does it perform regularization? Explain with a suitable example. (10)

- (b) During training of a Transformer model, for the input “This is a cat”, check the information below, where symbols have their usual meanings. What will be the matrix input to the softmax layer? (25)

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Contd...Q. No. 7(b)

	X					W^Q				W^K				W^V			
This	2	-1	-3	4	-2	1	0	0	0	0	1	0	0	1	0	0	0
is	-1	3	2	-1	-3	0	1	0	0	0	0	0	1	0	1	0	0
a	3	-2	-1	-2	2	0	0	1	0	1	0	0	0	0	0	1	0
cat	-3	3	0	-1	1	0	0	0	1	0	0	1	0	0	0	0	1
						0	0	0	0	0	0	0	0	0	0	0	1

8. (a) In the CBOW formation of Word2Vec, consider the portion of the neural network between the hidden layer and the output layer in Fig. for Q. (8). (10)

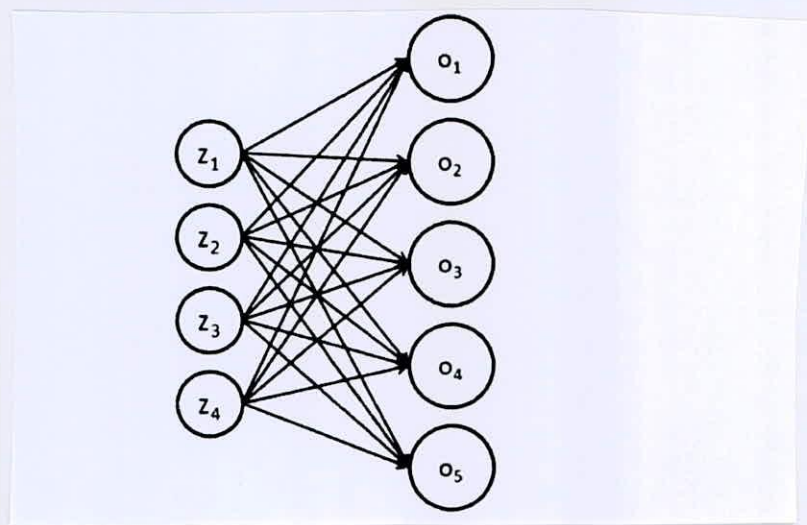


Figure for Q. No. 8(a).

The one-hot encodings of the words in the vocabulary are as follows:

	Tiger	Horse	Cow	Cat	Dog
o_1	1	0	0	0	0
o_2	0	1	0	0	0
o_3	0	0	1	0	0
o_4	0	0	0	1	0
o_5	0	0	0	0	1

The weights of the links between the hidden layer and output layer are as follows, where the value in cell $z_i o_j$ represents the weight of the link from hidden node z_i to output node o_j :

Contd P/4

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Contd...Q. No. 8

	$\mathbf{o}_1$	$\mathbf{o}_2$	$\mathbf{o}_3$	$\mathbf{o}_4$	$\mathbf{o}_5$
$\mathbf{z}_1$	2	4	6	1	5
$\mathbf{z}_2$	-1	3	4	2	-5
$\mathbf{z}_3$	3	-2	2	4	4
$\mathbf{z}_4$	4	5	1	-3	2

Write down the Word2Vec embedding of each word in the vocabulary.

(b) What is a GAN? Describe its various components. Deduce its loss functions.

(10)

(c) A Variational Auto Encoder (VAE) has learnt μ and σ vectors as follows:

$\mu = [1 \ 5 \ 2 \ 8 \ 9 \ 7]^T$, $\sigma = [2 \ 3 \ 2 \ 1 \ 5 \ 3]^T$, A sample drawn from a multivariate normal distribution $N([2 \ 1 \ 1 \ 3 \ -3 \ 2]^T, \text{diag}[4, 4, 9, -4, -16, 1])$ is $[1 \ 2 \ -1 \ -2 \ 3 \ 2]^T$. Using this information, calculate the corresponding latent representation (z) of the sample generated by the VAE.

(15)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) What is ensemble learning? Explain bagging and boosting. (10)
 (b) Deduce the learning rules for the weights in univariate linear regression. (10)
 (c) Write short notes on: (15)
 - (i) Supervised vs. unsupervised learning
 - (ii) Validation test vs. cross-validation test
 - (iii) Bias-variance trade off

2. (a) What do you understand by regularization? What are L_1 and L_2 regularizations? Why does L_1 regularization tend to produce a sparse model? (5+5+5=15)
 (b) What is a Gaussian mixture model? Write down and explain the equation for the probability of a data point under this model. How can you learn the parameters of this model using the EM algorithm? (15)
 (c) What is a MAP hypothesis? What are its advantages and disadvantages over Bayesian learning? (5)

3. (a) What do you understand by discriminative and generative models? (5)
 (b) What is a Markov blanket of a node in a Bayesian Network? State the independence property related to it. (5)
 (c) Take a look at the Bayesian Network (BN) structure given in the Figure for Q. 3(c), and consider the various survey results as specified below. (10+15=25)
 - 2% of the population in the United States has ocular hypertension.
 - The global diabetes prevalence in 2019 is estimated to be 9.3%.
 - Table 1 summarizes another survey result on Diabetic and Non-diabetic people. The number of data points is mentioned directly in a cell, or inside parenthesis, as appropriate. As an example, from Table 1, we can see that 30 out of 100 diabetic patients encounter loss of weight. Of 50 patients having diabetes as well as ocular hypertension, 40 had glaucoma.
 - Table 2 gives the Conditional Probability Table (CPT) of change in vision.

Table 1: Results of survey

	Loss of Weight	Retinopathy	Glaucoma	Ocular Hypertension
Diabetic (100)	30	35	40	Yes (50)
			12	No (50)
Non- Diabetic (100)	15	10	30	Yes (50)
			4	No (50)

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Contd... Q. No.3(c)

Table 2: CPT of Change in Vision

G	R	P(C=true G,R)
t	t	0.90
t	f	0.60
f	t	0.65
f	f	0.20

- (i) Construct the CPT table for each of the remaining nodes of the network.
- (ii) A patient came to the hospital with complaint of change in vision. In the preliminary screening room, his weight was measured to be 60.5 Kg. Hearing the weight reading, the patient remarked that his weight has been around the 60 Kg mark for many years now. His ocular pressure check also ruled out ocular hypertension. What is the posterior probability that the person has diabetes?

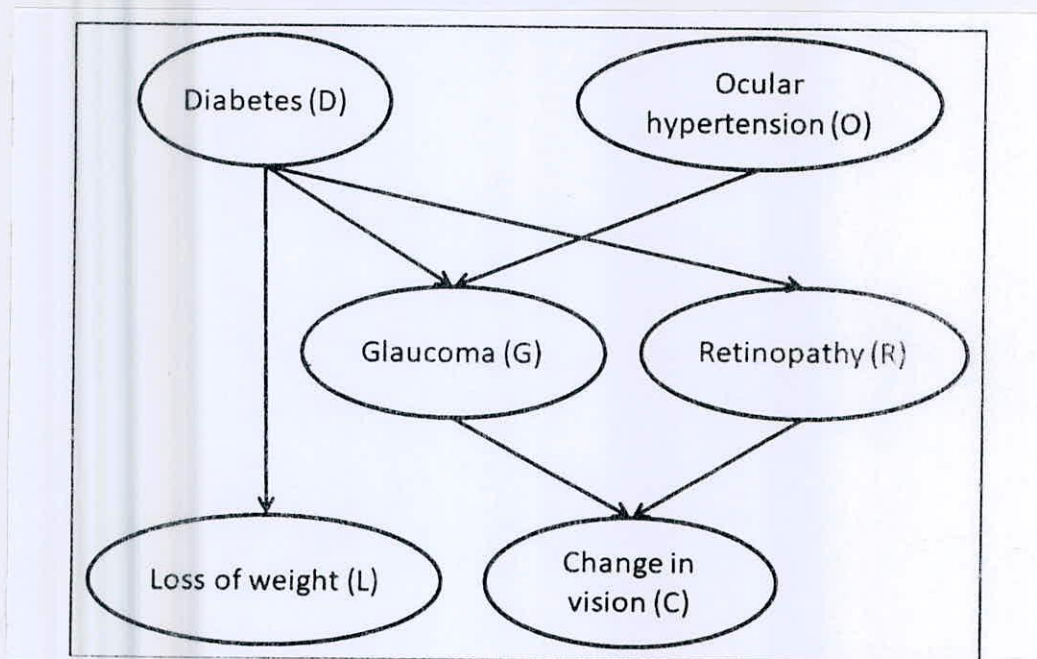


Figure for Q. 3(c)

4. (a) What do you understand by Markov Decision Process (MDP) and Reinforcement Learning (RL)? How are they different? Explain with an example. (10)
- (b) What is *Additive discounted reward*? How can you justify it? Show that even in the case of infinite horizon, additive discounted reward scheme results in bounded rewards. (10)
- (c) What is Q-function? Deduce its Bellman equation. (10)
- (d) Write down and explain the update equation for Temporal Difference Learning. (5)

CSE 471**SECTION – B**

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) What is your favourite application of deep learning? Briefly discuss about a model that can be used in this area. (10)
- (b) You have passed a vector through a softmax function and computed the cross-entropy loss with respect to the reference labels. Derive the gradient of the loss with respect to the softmaxed vector. (10)
- (c) Compute the derivatives of the function (15)

$$f(w, b, x) = \frac{1}{1 + \exp(w_0 x_0 + w_1 x_1 + b)}$$
 with respect to $(w_0, w_1, b) = (2.00, -3.00, -3.00)$ at $(x_0, x_1) = (0.40, -0.60)$, where w_0, w_1, b are model parameters and x_0, x_1 are the inputs.
6. (a) How do the two methods for learning word vectors (1) ‘continuous bag of words’ and (2) ‘skip-gram’ differ from one another? Can they be made more efficient? (10)
- (b) How does Faster-RCNN improve over Fast-RCNN? Why do RCNN models need pre-trained image classification models for object detection? (10)
- (c) Draw the computation graphs of an (1) Inception block and a (2) ResNeXt block. (Think of convolutions as black-box functions)
- An inception block has multiple convolutions (e.g., 1*1, 3*3, 5*5) and pooling (e.g., 3*3) and a ResNeXt block has multiple identical sequence of convolutions (each consisting of 1*1, followed by 3*3, followed by 1*1) after the input. (15)
7. (a) Explain the working mechanism of a stacked bidirectional RNN. (10)
- (b) Can a sequence-to-sequence Transformer network be used to copy words from the source texts? Explain how? (10)
- (c) You want to integrate the attention mechanism with an LSTM network. Your objective is to attend to both the cell state and the hidden state in parallel. How would you modify the model architecture? (You do not have to explain the internal working mechanisms of an LSTM) (15)
8. (a) How does Q-learning differ from policy gradient? Give an example of an RL method that can address their limitations. (10)
- (b) Name three model compression methods. Which method will you use to compress a model without reducing the number of parameters? Defend your position. (10)

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(c) Suppose you are using the Adam optimizer to train a model. While analyzing the loss landscape, you notice that the optimizer is not showing down at local minima.

Propose a modification to the optimizer to solve the issue.

(15)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t$$

An implementation of Adam is given above for your convenience.

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-4/T-2 B. Sc. Engineering Examinations (January 2020 Semester)

Sub: **CSE 471** (Machine Learning)

Full Marks: 180 Section Marks: 90 Time: 2 Hours (Sections A + B)

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – AThere are **FOUR** questions in this section. Answer any **THREE**.

No.	Questions	Marks
1.	a) Draw the computational graph of the function $f = \frac{1}{1 + e^{-[w_2 \ w_1 \ w_0] \begin{bmatrix} x_2 \\ x_1 \\ x_0 \end{bmatrix}}}$ For input $[w_2 \ w_1 \ w_0] = [-3 \ 4 \ -3]$ and $\begin{bmatrix} x_2 \\ x_1 \\ x_0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$, b) Compute the value of each intermediate node using forward propagation. c) Compute the output gradient with respect to each intermediate node using back propagation.	10+10+10
2.	a) State the difference between Q-function and policy-gradient based deep reinforcement learning architecture. Describe a scenario where we should prefer Q-function based architecture over policy-gradient based architecture. Explain your reason. b) Define vanishing gradient problem in the context of recurrent neural network (RNN). Between the following two sentences, which one will be more difficult to encode accurately using RNN. Explain your reason. Sentence 1: <i>The boy who played football yesterday on the school yard injured himself.</i> Sentence 2: <i>The boy injured himself yesterday during playing football on the school yard.</i>	5+5+5 5+5+5
3.	a) What are the steps of a single update of RMSProp optimization algorithm with Nesterov momentum? Describe the steps that should be modified to incorporate regular momentum instead of Nesterov. b) Below, you are given the architecture of a convolutional neural network. Write down the number of trainable parameters for each layer. The definition of each layer is given at the start. Definition: INPUT: $[W \times H \times D]$ = Input layer of dimension $W \times H \times D$ CONV_L_D: $[W \times H \times D]$ = Convolutional layer with total D filters of dimension $L \times L$ each producing filtered output of dimension $W \times H$ POOL_L: $[W \times H \times D]$ = Pooling layer with window dimension $L \times L$, stride L and output dimension $W \times H \times D$ FLATTEN: $[WHD]$ = Flatten input dimension $W \times H \times D$ to output dimension $1 \times WHD$ FC: $[D]$ = Fully connected layer with output dimension D	10+5 15

	Architecture: INPUT: $[32 \times 32 \times 3]$ CONV_3_64: $[32 \times 32 \times 64]$ CONV_3_64: $[32 \times 32 \times 64]$ POOL_2: $[16 \times 16 \times 64]$ CONV_3_8: $[16 \times 16 \times 8]$ CONV_3_8: $[16 \times 16 \times 8]$ POOL_2: $[8 \times 8 \times 8]$ FLATTEN: $[512]$ FC: $[32]$ FC: $[10]$	
4.	a) How do we solve the problem of backpropagating gradients through sampling layer during training of variational autoencoder (VAE)? Define the two parts of the empirical loss function for VAE training and describe the role of each part.	5+10
	b) What does the latent variable represent in a mixture model? Write E-step and M-step of the EM algorithm for a mixture of n Gaussian distribution.	5+5+5

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-4/T-2 B. Sc. Engineering Examinations 2018-19

Sub: **CSE 471** (Machine Learning)

Full Marks: 180 Section Marks: 90 Time: 2 Hours (Sections A + B)

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – BThere are **FOUR** questions in this section. Answer any **THREE**.

5. (a) "Maximizing the projection variance on a principal axis is equivalent to minimizing the errors due to projection"-Justify the clause in the context of Principal Component Analysis (PCA). (10)
- (b) What is eigenvector and what is eigenvalue? How are they related to the PCA? How does covariance matrix help to determine the principal components? Briefly explain. (20)
6. (a) How does Boosting help to get a better classifier? Briefly explain. (10)
- (b) Determine the decision stump (one-level decision tree) based on the following constraints and data (Table 6.1) using information gain measure. (20)

Constraints:

- i. Exclude the row of Table 6.1 with serial no. n , where

$$n = (3 * (\text{last three digit of your student id})) \% 10$$

For example, if the last three-digit of your student id is 123, then $n = (3 * 123) \% 10 = 9$. In that case, Row no. 9 will be excluded, and all other rows will be considered.

- ii. Consider A and B as input features, and C as class label. If there is a tie in the leaf node, then consider '+' as the class label.

Serial No.	A	B	C
0	F	F	-
1	T	F	+
2	T	T	+
3	T	T	+
4	T	F	-
5	T	T	+
6	F	F	-
7	F	F	-
8	T	T	-
9	T	F	-

Table 6.1: Data for question 6(b)

7. (a) What is cost function? What problem does gradient descent face for a non-convex cost function? (5+5)
- (b) Write down the main problem for each of the followings: (4 x 2 $\frac{1}{2}$)
- Too simple hypothesis
 - Too complex hypothesis
 - Batch gradient descent
 - Stochastic gradient descent
- (c) For which type of datasets it is useful to apply k -fold cross validation? Why may it work better than traditional 2-split (train and validation/test) approach in these datasets? Briefly explain. (10)

8. (a) How do we get sigmoid function from *logit* function? Why is sigmoid function important in logistic regression? (5+5)
- (b) Although kNN generally requires structured input, the number of words in a text document can vary a lot. How do we apply kNN for such unstructured and variable-length text documents? Why is Inverted Document Frequency (IDF) important in case of Document Classification? (5+5)
- (c) What are the fundamental assumptions behind Naive Bayes algorithm? What is the difference between equal-interval binning and equal-frequency binning? (5+5)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Which variant of stochastic gradient descent algorithm with adaptive learning rate does use interim updates? Write its gradient accumulation and velocity update steps with corresponding equation. (5+5+5)

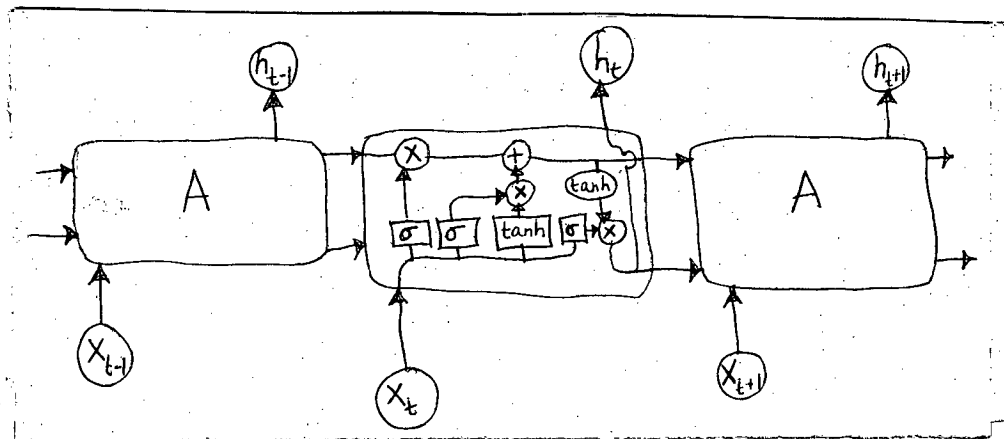
- (b) A 3-node hidden layer word2vec model has been trained over a 5-word unique vocabulary corpus. The weight matrix is given below. (10+10)

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 2 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \\ 6 & 5 & 4 \\ 9 & 9 & 9 \end{bmatrix}$$

What are the one-hot encoding of two most similar words and two most dissimilar words in the corpus?

2. (a) A schematic figure of a recurrent LSTM unit is given below. Redraw the unit which (5+5+5)

- (i) produces output between 0 to 1
- (ii) has multiplicative cell state update
- (iii) uses tanh forget gate



- (b) Here you are given the architecture of a convolutional neural network. Calculate the number of trainable parameters for each layer. The definition of each layer is given below. (20)

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Contd... Q. No. 2(b)

- (i) INPUT: $[W \times H \times D]$ = Input layer of dimension $W \times H \times D$
- (ii) CONV_L_D: $[W \times H \times D]$ = Convolutional layer with total D filters of dimension $L \times L$ each producing filtered output of dimension $W \times H$
- (iii) POOL_L: $[W \times H \times D]$ = Pooling layer with window dimension $L \times L$, stride L and output dimension $W \times H \times D$
- (iv) FLATTEN: $[WHD]$ = Flatten input dimension $W \times H \times D$ to output dimension $1 \times WHD$
- (v) FC: $[D]$ = Fully connected layer with output dimension D

Architecture:

The network architecture is given below:

INPUT: $[32 \times 32 \times 3]$
CONV_3_64: $[32 \times 32 \times 64]$
CONV_3_64: $[32 \times 32 \times 64]$
POOL_2: $[16 \times 16 \times 64]$
CONV_3_8: $[16 \times 16 \times 8]$
CONV_3_8: $[16 \times 16 \times 8]$
POOL_2: $[8 \times 8 \times 8]$
FLATTEN: $[512]$
FC: $[32]$
FC: $[2]$

3. (a) Write the update equations of row latent factor and column latent factor for alternating least square matrix factorization algorithm without any regularization. Define each quantity in your equations. (10+10)

(b) Consider 3 data points in the 2-d space: $(-1,-1)$, $(0,0)$, $(1,1)$. Find the first principal component (write down the actual vector) from their graph plot (no numerical eigen decomposition should be necessary). Calculate its corresponding eigen value. (5+10)

- 4 Let, a vector $\mathbf{x}$ with dimension D can be generated from any one of the k independent Gaussian distributions with identity co-variance matrix. The probability of selection of Gaussian distribution i is w_i where $\sum_{i=1}^k w_i = 1$ and the probability of generation of

sample $\mathbf{x}_j$ is given as
$$P(\mathbf{x}_j | \boldsymbol{\mu}_i, \mathbf{I}) = \frac{1}{\sqrt{(2\pi)^D}} e^{\left(-\frac{1}{2}(\mathbf{x}_j - \boldsymbol{\mu}_i)^T (\mathbf{x}_j - \boldsymbol{\mu}_i)\right)}$$

For this particular model answer the following questions

- (a) Write a generative model for above scenario. (5)

Contd P/3

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Contd... Q. No. 4

- (b) If $\mathbf{x}_j$ denotes a position vector within 3-dimensional co-ordinate system and there are 4 mixture components, calculate the total number of trainable parameter. (10)
- (c) Write the four steps of the expectation-maximization algorithm to estimate trainable parameters for above model. (20)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Define the least squares linear regression model and ridge regression model. While finding the optimized weight vector $\mathbf{w}$, what is the problem of these models when the dimension $\mathbf{D}$ is very large? How can we solve this? Explain with necessary mathematical explanations. (13)
- (b) Write short notes on: (12)
- (i) Locality sensitive hashing
 - (ii) Locally weighted regression
 - (iii) k-d tree
- (c) Explain with example - "Nearest neighbor approaches are sensitive to the curse of dimensionality". Explain two ways to solve this problem. (10)
6. (a) Define entropy and information gain. Given the following training examples for calculating risk of loan credit, find a decision tree using information gain heuristics. (18)

No.	History	Debt	Collateral	Income	Risk
1	bad	High	none	\$0-15K	high
2	unknown	High	none	\$15-35K	high
3	unknown	Low	none	\$15-35K	mod
4	unknown	Low	none	\$0-15K	high
5	unknown	Low	none	over \$35K	low
6	unknown	Low	adequate	over \$35K	low
7	unknown	Low	none	\$0 to 15K	high
8	bad	Low	adequate	over \$35K	mod
9	good	Low	none	over \$35K	low
10	good	High	adequate	over \$35K	low
11	good	High	none	\$0 to \$15K	high
12	good	High	none	\$15 to \$35K	mod
13	good	High	none	over \$35K	low
14	bad	High	none	\$15 to \$35K	high

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Contd... Q. No. 6

(b) Consider the following plot of performance of a decision tree in Figure 6(b).

(10)

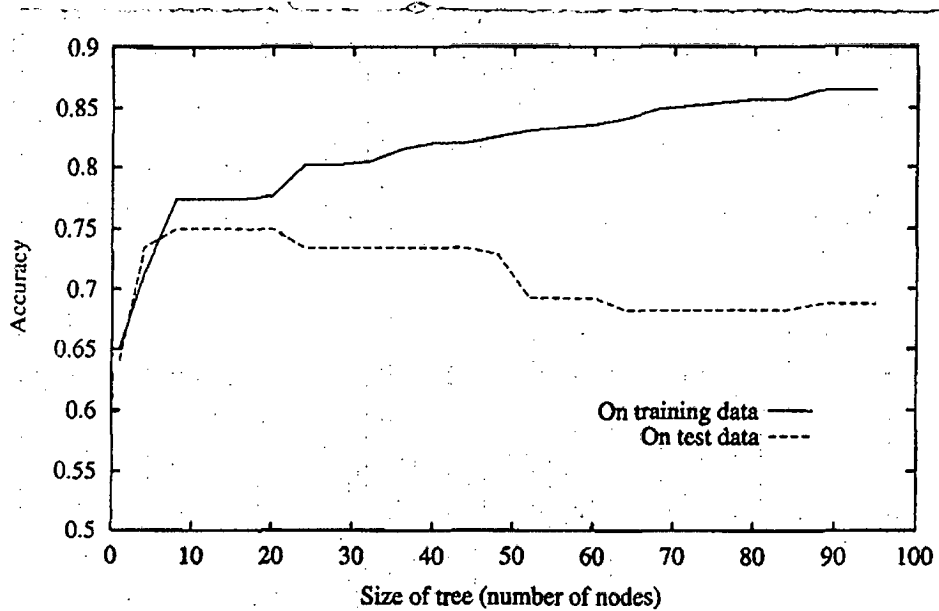


Figure 6(b)

The horizontal axis of this plot indicates the total number of nodes in the decision tree, as the tree is being constructed. The vertical axis indicates the accuracy of predictions made by the tree. The solid line shows the accuracy of the decision tree over the training examples, whereas the broken line shows accuracy measured over an independent set of test example (not included in the training set).

(i) Why did the decision tree behave in this way?

(ii) How can one get rid of this behavior?

(c) What is underfitting of data? How can this be avoided?

(7)

7. (a) Define weak and strong classifiers. How does AdaBoost algorithm assign weights to different classifiers so that combined classification error is minimized? Does AdaBoost algorithm cause overfitting of data? Justify your answer.

(5+9+5)

(b) What is the significance of generalization in the context of machine learning algorithms? Differentiate between supervised learning and semi supervised learning.

(4+4)

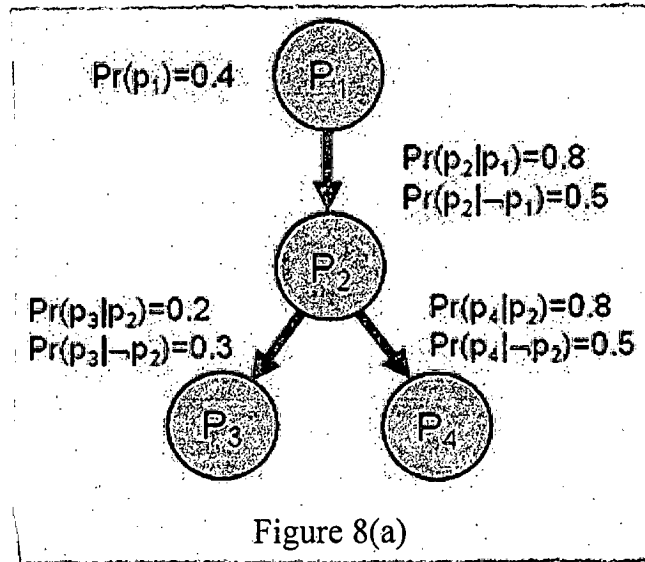
(c) Explain bagging and stacking in context of ensemble learning.

(8)

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8. (a) Consider the following Bayesian Network in Figure 8(a):

(20)



Calculate marginal and conditional probabilities $\Pr(\neg p_3)$, $\Pr(p_2 | \neg p_3)$ and $\Pr(p_1 | p_2, \neg p_3)$ using the method of **inference by enumeration**.

- (b) Consider you have to develop a model which identifies Nobel laureates from average people. In the given dataset, there are 80 million people and of them 193 (confirmed) were Nobel laureates. Now, you decided not to do any work and simply label "Not Nobel laureate" to every people. You feel happy about your model "accuracy" which is 99.9998%!!

(12)

What is wrong with "accuracy" for this scenario? Describe three metrics which can be helpful for pointing out the pitfalls in this type of scenario. Explain with example.

- (c) Define **peeking** in context of supervised learning.

(3)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) What is the objective function to optimize when finding the first principal component of a non-square data matrix? How does this objective function capture the direction of maximum variance in data? (5+5)
(b) Derive the loss function for sparse matrix factorization. Write the pseudocode of alternating least square algorithm handling sparsity. (10+5)
(c) When should we not use nearest-neighbor non-parametric models? How can we handle such situation? (5+5)
2. (a) Probability of a person having no family history of diabetes (no history) is 0.70 (5+10)
Probability of having Diabetes when the corresponding person
has no history is 0.25
has history is 0.65
Probability of having Hyperglycemia when a person has
Diabetes is 0.85
no Diabetes is 0.20
Probability of having Hypertension when a person has
Diabetes is 0.60
no Diabetes is 0.40
You can safely ignore all other possible dependencies. From the above information, draw and annotate the corresponding Bayesian network with conditional probability tables. For this Bayesian network, infer the marginal distribution of $P(\text{Diabetes, Hyperglycemia})$ when we have evidence that the person has a family history of Diabetes and is diagnosed with Hypertension.
(b) For the Bayesian network in question 2(a) with same evidence variables, find the conditional distributions of other two unknown variables. Write the pseudocode of a Gibb's sampler for the inference of marginal distribution of these two unknown variables. (10+10)

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3. (a) What is the latent variable in a mixture model? Write the generative model of a mixture of generic distributions P . (5+5)
- (b) Derive analytical solution of the M-step for a univariate Gaussian mixture model (GMM). (15)
- (c) Write the expected complete log-likelihood for a GMM with a Dirichlet prior over the mixture proportions. (10)
4. (a) What are the four inference tasks in the context of state-space model? Give their mathematical formulations. (4+6)
- (b) Draw an LSTM with input, forget and output gate. How these gates control information flow through time? (10+5)
- (c) Why do we need pooling layer in a deep convolutional network? Which two parameters do you need for a max-pooling layer? (5+5)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) What is the key difference between traditional algorithms and machine learning algorithms? Briefly explain five real world applications of machine learning algorithms. (8)
- (b) What is the significance of generalization in the context of machine learning algorithms? Differentiate between supervised learning and unsupervised learning. (5+4=9)
- (c) Briefly explain the Binomial and multivariate Gaussian probability distributions. (8)
- (d) Briefly explain the maximum likelihood estimation technique. (10)
6. (a) How does the cross-validation technique evaluate machine learning algorithms? Explain. (8)
- (b) What is stacking in the context of ensemble learning? (5)
- (c) Explain the bagging approach in ensemble learning. What is the key difference between random forest bagging and traditional bagging approaches? (8+4=12)
- (d) From the confusion matrix given below, calculate the following: sensitivity, specificity, precision, recall, and F1 score. (10)

		Predicted	
Actual		Buyer	Non-Buyer
	Buyer	516	25
	Non-Buyer	10	725

Contd P/3

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7. (a) Define entropy and information gain. Build a decision tree using ID3 algorithm from the following data. (6+14=20)

Day	Weather	Hunger	Location	Bought Food
1	Hot	Hungry	Home	No
2	Hot	Hungry	Not Home	Yes
3	Cold	Not Hungry	Home	No
4	Hot	Not Hungry	Not Home	No
5	Cold	Hungry	Home	No
6	Cold	Not Hungry	Not Home	No
7	Hot	Hungry	Not Home	Yes
8	Cold	Hungry	Not Home	Yes
9	Cold	Not Hungry	Home	No
10	Hot	Not Hungry	Home	No

- (b) Describe an approach to solve the overfitting problem in the context of decision trees. (7)
- (c) Briefly explain the kernel density estimation approach. (8)
8. (a) Define the least squares linear regression model. Explain how we can obtain the optimized weight vector for this model without considering regulation. (10)
- (b) What is the importance of regularization in risk minimization? Briefly explain the ridge regression model. (3+7=10)
- (c) What is the advantage of online learning over offline learning? (5)
- (d) Estimate the parameter θ for a Bernoulli distribution using the maximum a posteriori estimation approach where θ denotes the probability of head. Assume that the prior distribution of θ is an uniform distribution over the interval $(0, 1)$. (10)
-

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Draw a complete figure containing all the components of learning. Describe each component in short. (15)
- (b) Discuss about the complexities of the unknown target function f and hypothesis set $\mathcal{H}$ in case of learning. How does the noise affect the unknown target function? (10+5=15)
- (c) What is the chromosomal representation technique in an evolutionary programming? What are the possible mutation operators to create an offspring in evolutionary programming? (2.5+2.5=5)
2. (a) Briefly explain the following concepts with the necessary examples. (4+4+4+3=15)
 - (i) Growth function, (ii) Break point, (iii) VC-dimension and (iv) Shattering.
- (b) Briefly describe chromosomal representation technique, different generic operators, parent selection technique and survivor selection technique of generic programming with necessary examples. (3+8+2+2=15)
- (c) Why is the VC bound very loose? (5)
3. (a) Discuss the effect of model complexity on bias and variance. How do the learning curves help to decide whether more data may help very much for better learning? (8+7=15)
- (b) How is the bin example similar to learning? Write down the Hoeffding inequality for a single hypothesis. How can it be extended for multiple hypotheses? (4+3+8=15)
- (c) What has made cosine similarity different from the other distance measures for k-nearest neighbor selection? What is the importance of IDF in cosine similarity? (2+3=5)
4. (a) How do we get 'VC-inequality' from 'Hoeffding inequality for multiple hypotheses'? Briefly describe with pictorial proof and other related figures. (15)
- (b) How do the generative learning algorithms differ from the discriminative learning algorithms? Where should we use Gaussian discriminant analysis and where should we use Naïve Bayes? What is the strong assumption of Naïve Bayes? Why Laplace smoothing factor is necessary for Naïve Bayes? (4+4+2+5=15)
- (c) How does generic programming (GP) differ from the other evolutionary algorithms? (5)

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Derive the Generalized Linear Model (GML) for the classification of an input dataset into k classes. Assume necessary probabilistic distribution for the data. (15)
- (b) Write down the Markov Decision Process based Q-learning (one type of reinforcement) algorithm. How does Q-learning for deterministic Markov decision processes converge? (12+8)
6. (a) Explain batch gradient descent and stochastic gradient descent based linear regression algorithms. When is stochastic gradient descent often preferred over batch gradient descent? (10+5)
- (b) Show that under certain probabilistic assumptions on the data, least-squares regression corresponds to finding the maximum likelihood estimate of parameter, θ . (15)
- (c) Define Inductive and Transductive learning? (5)
7. (a) What is Bagging? How is it related to bias-variance tradeoff? (5+5)
- (b) Draw a block diagram of AdaBoost ensemble method by showing its different steps? How do you choose weights for different hypotheses in AdaBoost? (10+5)
- (c) What is the main difference between bagging based decision tree and random forest based decision tree? (10)
8. (a) Consider data with two Boolean attributes (A, B). The data set consists of 203 examples labeled with + and – outputs. Based on the information gain concept, draw the decision tree. (15)
- $\langle A=0, B=0 \rangle, -$: 50 examples
 $\langle A=0, B=1 \rangle, -$: 50 examples
 $\langle A=1, B=0 \rangle, -$: 3 examples
 $\langle A=1, B=1 \rangle, +$: 100 examples
- (b) How should you modify the information gain based attributes selection method to handle attributes with *many* values and attributes with *missing* values? (10)
- (c) Why does overfitting occur in a decision tree based classification? What are the remedies to avoid overfitting? (5+5)
-

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Define the following basic principles of Machine Learning: (i) Occam's razor, (ii) sampling bias, and (iii) data snooping. (9)
- (b) Show the major components of learning and their relationship using a block diagram. (10)
- (c) For the linear support vector machine (SVM), derive the function with necessary constraints that we need to minimize the margin. (10)
- (d) What is Hoeffding's inequality in learning theory? (6)

2. (a) What function do you maximize in Markov decision process based re-enforcement learning? (5)
- (b) Derive Bellman's equation for the optimal value function. How do you find the optimal policy from the derived equation? (10+5)
- (c) Show that the expected out of sample error can be expressed as the summation of bias and variance. (10)
- (d) What's the purpose of a soft margin SVM? (5)

3. (a) Show that $K(x, x') = (1 + x^T x')^2$ is a valid kernel for SVM, where x is a 2-dimensional data. (10)
- (b) Define the growth function and break point. What is the relationship between the growth function and a break point? (5+5)
- (c) How do you compute k principal components of the data? When the data dimension is big, what strategy should you take to find principal components? (10+5)

4. (a) Let $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$ be a given set of input and $z^{(i)}$'s be the random variables that indicate which of the k Gaussians each $x^{(i)}$ come from. Write down the steps of Expectation Maximization (EM) algorithm for density estimation of unknown latent variables z in a mixture of Gaussians model. If the distribution does not follow Gaussians, how should you modify the above algorithm using Jensen's inequality? (10+10)
- (b) Show that the VC dimension, d_{vc} of perceptrons in a d dimensional dataspace is $d+1$. (10)
- (c) Give a real-world learning example where a false accept should get a higher priority than a false reject. (5)

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Explain the reasons of choosing an ensemble method over a single classifier with a suitable example. (7)
- (b) Write the differences between bagging and boosting methods of ensemble learning? (7)
- (c) Define weak and strong classifiers. How does AdaBoost algorithm assign weights to different classifiers so that combined classification error is minimized? (4+11)
- (d) Does AdaBoost algorithm cause overfitting of data? Justify your answer. (6)
6. (a) The following table contains training examples to determine whether or not Mashrafe enjoys cricket. (18+5)

	Sky	Temperature	Humidity	Wind	Play Cricket
1	Sunny	42	High	Weak	No
2	Sunny	35	High	Strong	Yes
3	Overcast	44	High	Weak	No
4	Rain	20	Normal	Weak	No
5	Rain	12	Normal	Strong	No
6	Overcast	25	Normal	Strong	Yes
7	Overcast	32	Normal	Weak	Yes

In the table temperature is continuous valued attribute and the rests are discrete valued attribute.

- (i) Using ID3 algorithm construct a minimal decision tree that predicts whether or not Mashrafe enjoys cricket on a certain day. Show each step of your computation.
- (ii) Translate your decision tree in 6(a). (i) to a collection of rules.
- (b) Explain with examples how the following cases are handled in Decision Tree Learning Algorithm: (4+4=8)
- (i) Attributes with missing values
- (ii) Attributes with differing costs
- (c) Describe the impact of high learning rate on stochastic gradient descent search? (4)
7. (a) Define likelihood in terms of probabilistic interpretation. What is the advantage of measuring log likelihood over pure likelihood in case of logistic regression? (6)
- (b) What is overfitting of data? How overfitting of data can be avoided? (8)
- (c) What will be the change in update rule of Newton's method if it is used to minimize rather than maximize a function? (7)

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Contd ... Q. No. 7

- (d) Suppose, you want to build an email spam filter using a machine learning technique. Here, messages are only classified according to whether they are spam or non-spam email. After learning to do this depending on a training set, we want our email reader to automatically filter out the spam messages and perhaps place them in a separate mail folder. Which machine learning technique do you prefer to use? Model the email filter depending on your learning technique. (14)
8. (a) When does logistic regression always show better performance than Gaussian Discriminant Analysis? Explain the reason. (3+5=8)
- (b) Does a perception algorithm provide a unique classifier if the training examples are linearly separable? Justify your answer. (10)
- (c) What do you want a regressor to do? (5)
- (d) Briefly explain ruled post pruning and reduced error pruning? (12)
-

SECTION – AThere are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) Give the examples of three computer applications for which machine learning approaches seem appropriate and three for which they seem inappropriate. Write justifications for each of them. (15)
- (b) “The choices of training data and target concepts are two important components in designing a machine learning system” – Justify the statement with appropriate examples. (13)
- (c) Explain the facilities and pitfalls of Find-S algorithm. (7)
2. (a) Give the sequence of S and G boundary sets computed by the Candidate Elimination algorithm for the training examples shown in the following table. Can you come with the ideas for ordering the training examples to minimize the sum of the sizes of the intermediate S and G sets? (18)

Examp le	Sky	AirTem p	Humidit y	Wind	Water	Foreca st	EnjoySpo rt
1	Sunny	Warm	High	Strong	Cool	Change	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Warm	Same	Yes
4	Sunny	Warm	Normal	Strong	Warm	Same	Yes

- (b) Semi-supervised learning is suitable in many practical scenarios. Explain this fact with a suitable example. (7)
- (c) What are the differences between the t -test and Wilcoxon signed-rank test? Explain when the Wilcoxon signed-rank test is preferable than the t -test. (10)
3. (a) What is it a problem for a learner to learn from an imbalanced dataset? Explain two techniques to deal with this. (20)
- (b) Traditional classification techniques are not appropriate for stream data classification. Give some intuitive ideas how to modify such classifiers for stream data classification. (15)
4. (a) What is meant by PAC learnability? Show that

$$m \geq \frac{1}{\epsilon} (\ln |H| + \ln) \left(\frac{1}{\delta} \right)$$

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Contd ... Q. No. 4(a)

where m is the number of training example, H is the hypothesis spaces, ϵ is the error and δ is some threshold. What are the utilities of Vapnik-Chervonenkis (VC) dimension in machine learning?

(20)

(b) Give general expressions for the upper and lower one-sided $N\%$ confidence intervals for the difference in errors between two hypotheses tested on different samples of data.

(15)

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) Consider the Sequential-Covering algorithm shown in figure 5(a) for learning a disjunctive set of propositional rules.

```

SEQUENTIAL-COVERING(Target_attribute, Attributes, Examples, Threshold)
• Learned_rules ← {}
• Rule ← LEARN-ONE-RULE(Target_attribute, Attributes, Examples)
• while PERFORMANCE(Rule, Examples) > Threshold, do
    • Learned_rules ← Learned_rules + Rule
    • Examples ← Examples – {examples correctly classified by Rule}
    • Rule ← LEARN-ONE-RULE(Target_attribute, Attributes, Examples)
• Learned_rules ← sort Learned_rules accord to PERFORMANCE over Examples
• return Learned_rules
    
```

Figure- 5(a)

Here **LEARN-ONE-RULE** must return a single rule that covers at least some of the Examples. **PERFORMANCE** is a user-provided subroutine to evaluate rule quality. (10+3+7=20)

(i) Implement **LEARN-ONE-RULE** as a greedy general-to-specific search which begins by considering the most general rule precondition, then greedily adding the attribute test that most improves rule performance measured over the training examples. You have to write the corresponding pseudo-code only.

(ii) What is the main limitation of the implementation asked in (i)?

(iii) How can you extend the implementation asked in (i) to reduce the limitation identified in (ii)?

(b) Discuss the advantages of first-order representations over propositional (variable-free) representation with a suitable example.

(5)

(c) Consider learning rules to predict the target literal *GrandDaughter* (x, y), where the other predicates used to describe examples are *Father* and *Female*. The general-to-specific search in **FOIL** algorithm begins with the most general rule.

GrandDaughter (x, y) ←

Which asserts that *GrandDaughter* (x, y) is true of any x and y . List the candidate literals generated by **FOIL** to specialize this initial rule.

(10)

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6. (a) Consider the deterministic grid world shown in figure 6(a) with the absorbing goal-state G. Here the immediate rewards are 10 for the labeled transitions and 0 for all unlabeled transitions.

(10+10+5+5=30)

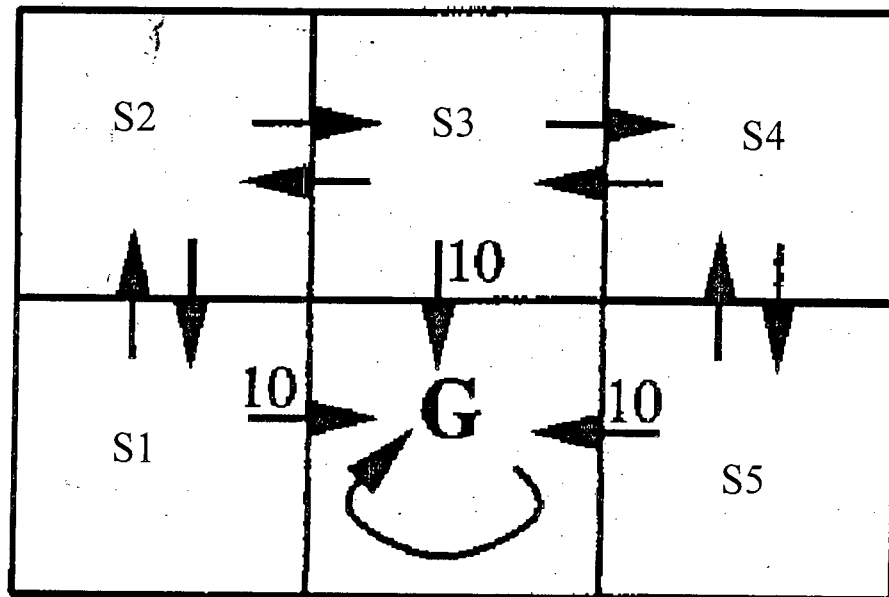


Figure - 6(a)

Q learning algorithm

For each s, a initialize the table entry $\hat{Q}(s, a)$ to zero.

Observe the current state s

Do forever:

- Select an action a and execute it
- Receive immediate reward r
- Observe the new state s'
- Update the table entry for $\hat{Q}(s, a)$ as follows:

$$\hat{Q}(s, a) \leftarrow r + \gamma \max_{a'} \hat{Q}(s', a')$$

- $s \leftarrow s'$

Figure - 6(b)

- (i) Give the V^* value of every state in this grid world. Give the $Q(s, a)$ value for every transition. Finally, show an optimal policy. Use $\gamma = 0.9$.
- (ii) Now consider applying the **Q Learning algorithm** shown in figure 6(b) to this grid world, assuming the table of Q values is initialized to zero. Assume the agent begins in the bottom left grid square and then travels clockwise around the perimeter of the grid until it reaches the absorbing goal state, completing the first training episode. Describe which Q values are modified as a result of this episode, and give their revised values. Answer the question again assuming the agent now performs a second identical episode.
- (iii) Mention the conditions under which the algorithm of figure 6(b) will converge toward a $\hat{Q}$ equal to the true Q function.

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Contd ... Q. No. 6(a)

(iv) The algorithm of figure 6(b) does not specify how actions are chosen by the agent. Suggest a probabilistic scheme with parameter $p > 0$ for selecting actions. Larger value of p should cause the agent to exploit what it has learned whereas small values of p should lead the agent to explore actions that have not been tried often before.

(b) **Information gain** measure is biased towards attributes having a large number of values – can this bias lead to any trouble? Explain with a suitable example. (5)

7. (a) Suppose you are given some data regarding coal resource in Bangladesh as shown in table 7(a). Here location of each area is described using its latitude and longitude. Each area has an associated probability of containing coal field. This probability is unknown for the last 3 areas. (5+20=25)

Area	Latitude	Longitude	Probability of containing coal field
Natore	24.5	89.1	0.50
Kendua	24.7	91.1	0.62
Dinajpur	25.7	88.5	0.90
Dhaka	23.8	90.5	0.40
Sylhet	24.9	91.92	0.95
khulna	22.7	89.6	0.8
Sreemangal	24.53	91.9	0.56
Jessore	23.3	89.4	0.7
Bogra	24.8	89.45	?
Narshingdi	24.1	90.7	?
Satkhira	22.73	89.2	?

Table- 7 (a)

(i) Mention the assumptions necessary to apply **k-nearest neighbor** algorithm for finding the unknown probabilities. The **k-nearest** neighbor algorithm for approximating a discrete-valued function $f: \mathcal{X} \rightarrow V$ is shown in figure 7(a).

(ii) Apply **2-nearest neighbor** algorithm for finding the unknown probabilities. You must weight the contribution of each neighbor according to its distance to the query point. The weight of a neighbor is the inverse of its distance to the query point. You must show all the steps of calculation.

Training algorithm:

- For each training example $\langle x, f(x) \rangle$, add the example to the list *training_examples*

Classification algorithm:

- Given a query instance x_q to be classified,
 - Let $x_1 \dots x_k$ denote the k instances from *training_examples* that are nearest to x_q
 - Return

$$\hat{f}(x_q) \leftarrow \operatorname{argmax}_{v \in V} \sum_{i=1}^k \delta(v, f(x_i))$$

where $\delta(a, b) = 1$ if $a = b$ and where $\delta(a, b) = 0$ otherwise.

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Contd ... Q. No. 7

(b) Both the k-nearest neighbors and locally weighted regression construct an explicit approximation of the target function over a local region surrounding the query point - are you agree with this statement? Provide justification. (5)

(c) Radial basis function networks can be seen as a blend of instance-based approaches and neural network approaches – justify. (5)

8. (a) Construct the decision tree using ID3 algorithm (shown in figure 8a) with information gain to select the best attribute at each step for the customer database given in table 8(a). Here the target attribute **buys_computer** has two distinct values (yes, no). You must show all the steps of calculation. (25)

age	income	student	credit_rating	buys_computer
less than 30	high	no	fair	no
less than 30	high	no	excellent	no
between 30 and 40	high	no	fair	yes
greater than 40	medium	no	fair	yes
greater than 40	low	yes	fair	yes
greater than 40	low	yes	excellent	no
between 30 and 40	low	yes	excellent	yes
less than 30	medium	no	fair	no
less than 30	low	yes	fair	yes
greater than 40	medium	yes	fair	yes
less than 30	medium	yes	excellent	yes
between 30 and 40	medium	no	excellent	yes
between 30 and 40	high	yes	fair	yes
greater than 40	medium	no	excellent	no

Table – 8 (a)

ID3(Examples, Target_attribute, Attributes)

Examples are the training examples. Target_attribute is the attribute whose value is to be predicted by the tree. Attributes is a list of other attributes that may be tested by the learned decision tree. Returns a decision tree that correctly classifies the given Examples.

- Create a *Root* node for the tree
- If all *Examples* are positive, Return the single-node tree *Root*, with label = +
- If all *Examples* are negative, Return the single-node tree *Root*, with label = -
- If *Attributes* is empty, Return the single-node tree *Root*, with label = most common value of *Target_attribute* in *Examples*
- Otherwise Begin
 - $A \leftarrow$ the attribute from *Attributes* that best* classifies *Examples*
 - The decision attribute for *Root* $\leftarrow A$
 - For each possible value, v_i , of A ,
 - Add a new tree branch below *Root*, corresponding to the test $A = v_i$
 - Let $Examples_{v_i}$ be the subset of *Examples* that have value v_i for A
 - If $Examples_{v_i}$ is empty
 - Then below this new branch add a leaf node with label = most common value of *Target_attribute* in *Examples*
 - Else below this new branch add the subtree
ID3($Examples_{v_i}$, *Target_attribute*, $Attributes - \{A\}$)
- End
- Return *Root*

Figure – 8 (a)

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Contd ... Q. No. 8

(b) Suppose you are experimenting with ID3 algorithm for learning which medical patients have a form of diabetes. To visualize the performance of your generated decision tree you have drawn the plot of figure 8(b).

(5+5=10)

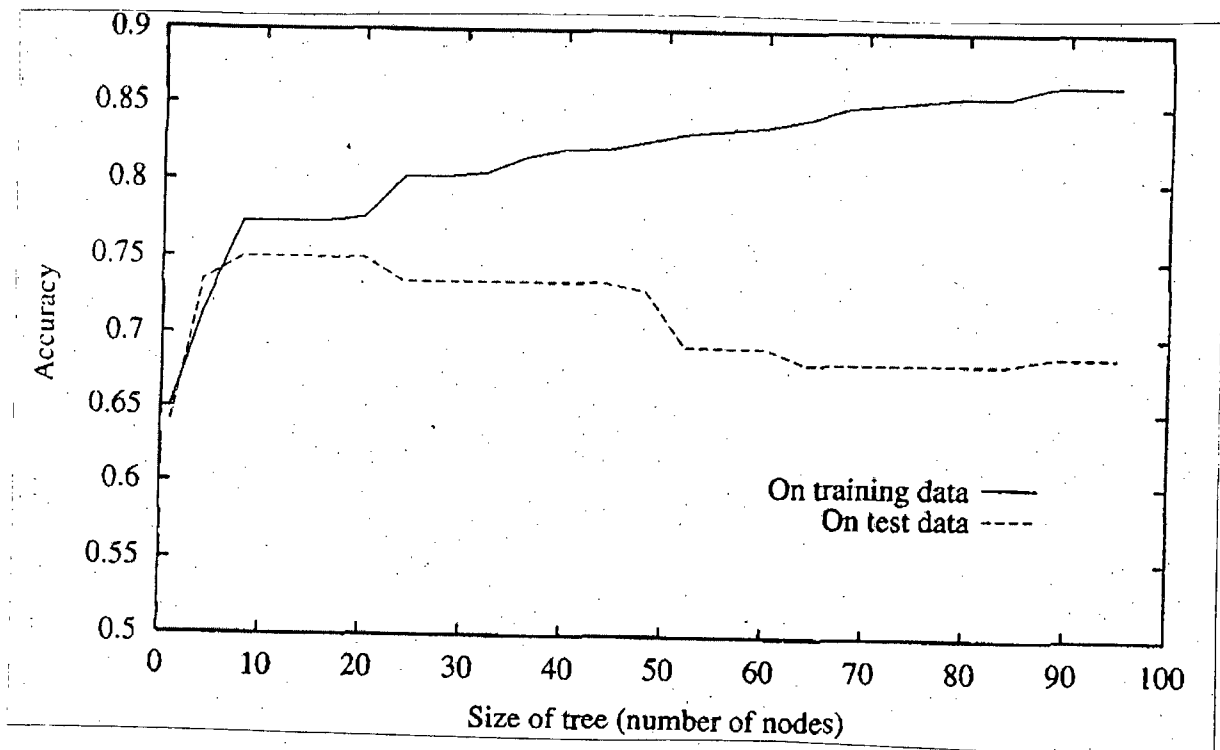


Figure - 8 (b)

The horizontal axis of this plot indicates the total number of nodes in the decision tree, as the tree is being constructed. The vertical axis indicates the accuracy of predictions made by the tree. The solid line shows the accuracy of the decision tree over the training examples, whereas the broken line shows accuracy measured over an independent set of test example (not included in the training set).

- (i) Can you guess why did your decision tree behave in this way? Mention two possible reasons.
- (ii) Can you suggest some approaches to get rid of this behavior?

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) What do you mean by PAC-learnability and shattering? Show the probability that the version space $VS_{H,D}$ is not ϵ -exhausted (with respect to c) is less than or equal to $|H|e^{-Em}$. Here the symbols have their usual meaning. (15)
- (b) Imagine yourself as an apprentice chef in a restaurant. Your first task is to figure out how to make a salad. The rules are supposedly simple: you are free to combine any of the ingredients as they are and you can also slice any of the ingredients into two distinct pieces before mixing them. Since you have learnt PAC learning theory, you wonder how much effort you would need to figure out the makeup in a salad.
 - (i) Suppose that a naive chef makes salads by combining only any of the ingredients. Given N available ingredients and that each salad made out of these constitutes a distinct hypothesis. How large would the hypothesis space be? Explain how you arrive at your answer. (3)
 - (ii) Suppose that a more experienced chef follows both rules when making a salad. How large is the hypothesis space now? Explain. (3)
 - (iii) An experienced chef decides to train you to discern the makeup of a salad by showing you the salad samples he has made. There are 6 available ingredients. If you would like to learn any salad at 0.01 error with probability 99%, how many sample salads would you want to see? Show your workings in clear steps. (5)
- (c) Consider a 2D space or x_1 - x_2 plane. What is the VC dimension of circles where points inside are labeled as 1's and those outside as 0's? Draw an example scenario with minimal number of points where these circles would fail to shatter the space. (3)
- (d) Explain case based reasoning using an appropriate example. (6)
2. (a) Discuss the key properties of k -nearest neighbor and locally weighted regression. (5)
- (b) Nearest neighbor approaches are sensitive to the *curse of dimensionality*. Explain at least two different methods to solve this problem. (11)
- (c) "The working procedure of the distance-weighted regression and the radial basis function network is very much similar" – Justify the statement. (9)
- (d) How does the radial basis function network differ from the feed-forward neural network? Why will the former network be more suitable than the later one for solving regression problems? (5)

Contd P/2

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Contd ... Q. No. 2

- (e) In k -nearest neighbors, the classification is achieved by majority vote in the vicinity of data. Suppose there are two classes of data, each of $n/2$ points overlapped to some extent in a 2-dimensional space. Describe what happens to the training error (using all available data) when the neighbor size k varies from n to 1. (5)
3. (a) With the help of a Pseudo code, describe the FOIL algorithm in detail for learning a set of first-order rules. (20)
- (b) What are the most essential differences between FOIL and Sequential-covering and Learn-one-rule algorithms? (5)
- (c) Explain when and why a rule based learning method will be suitable over the black box learning method. (5)
- (d) Explain how you can avoid the local optima problem of sequential covering algorithm. (5)
4. (a) Explain when and why semi-supervised learning will provide much worse performance than supervised learning. (5)
- (b) How can a machine learning approach be used for e-mail filtering? (5)
- (c) With analytical formulations, describe negative correlation learning. (7)
- (d) Write the analytical form of the error measure used for classification and regression problems. (2)
- (e) Why the use of learning inside an evolutionary process may help in finding better solutions of a given problem. (4)
- (f) Give the flow chart of genetic programming in solving a problem. (4)
- (g) Describe all the components of genetic algorithm for obtaining the weights of a neural network. (8)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Derive the expression for and interpret the significance of *information gain* heuristics used in ID3 algorithm. Given the following training examples, find a decision tree using information gain heuristics. Discuss the limitation of the tree generated. Propose a *gain ratio* based approach to remove the limitation of *information gain* based approach. What further issue does *gain ratio* face? (25)

Contd P/3

CSE 471**Contd ... Q. No. 5(a)**

Day	Outlook	Temperature	Humidity	Wind	Venue	Play Tennis
D1	Sunny	Hot	High	Weak	Dhaka	No
D2	Sunny	Hot	High	Strong	Chittagong	No
D3	Overcast	Hot	High	Weak	Rajshahi	Yes
D4	Rain	Mild	High	Weak	Khulna	Yes
D5	Rain	Cool	Normal	Weak	Sylhet	Yes
D6	Rain	Cool	Normal	Strong	Barisal	No
D7	Overcast	Cool	Normal	Strong	Rangpur	Yes
D8	Sunny	Mild	High	Weak	Bogra	No
D9	Sunny	Cool	Normal	Weak	Comilla	Yes
D10	Rain	Mild	Normal	Weak	N. Gonj	Yes
D11	Sunny	Mild	Normal	Strong	Rangpur	Yes
D12	Overcast	Mild	High	Strong	Dinajpur	Yes
D13	Overcast	Hot	Normal	Weak	Pabna	Yes
D14	Rain	Mild	High	Strong	Tangail	No

*Outlook, Temperature, Humidity Wind and Venue are attributes and Play tennis is the function to be learned.

(b) Describe a typical design cycle for a machine learning problem.

(10)

6. (a) State the CANDIDATE-ELIMINATION algorithm. Show that the version space generated by the CANDIDATE-ELIMINATION algorithm from the following set of examples is invariant to the ordering of the examples. Hints: Simulate the algorithm using the examples in the given and the reverse orders.

(23)

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Warm	Same	Yes
4	Sunny	Warm	High	Strong	Cool	Change	Yes

*EnjoySport is the target concept to be learned.

(b) Explain rule pruning in decision tree learning. What are the benefits of rule pruning over tree pruning?

(12)

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7. (a) Define *sample error*, *true error* and *confidence interval* for a discrete value hypothesis. With appropriate assumptions on *sample*, *hypothesis*, and the *distribution of samples*, relate the parameters of *sample error* and *true error* to a Binomial distribution. Show why the *sample error* is an unbiased estimator of the *true error*. Derive the expression for confidence interval when the sample size is sufficiently large. (23)
- (b) Define *version space*, *general boundary*, and *specific boundary* of a *version space*. How do the positive and negative examples influence the general and specific boundaries of the version space? (12)
8. (a) Briefly describe the attributes that training experience should have in a typical machine learning problem. (9)
- (b) Differentiate between ID3 and CANDIDATE-ELIMINATION algorithms in terms of the hypothesis spaces that they search, and the search strategies that they follow. From this, define preference bias and restriction bias. (11)
- (c) Derive the expression for "confidence interval" for the estimation of difference of error of two learning algorithms when the sample size is limited. (15)
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L-4/T-2/CSE

Date : 18/12/2012

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-4/T-2 B. Sc. Engineering Examinations 2010-2011

Sub : **CSE 471** (Machine Learning)

Full Marks : 210

Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Discuss three computer applications for which machine learning approaches seem appropriate and three for which they seem inappropriate. (7)
- (b) The size of decision trees is the crucial factor for learnability by ID3 and its variants. Unfortunately, not all simple Boolean functions can be expressed by small decision trees. Suppose that a concept c is merely a disjunction of some attribute values. (For example, $c = x_1 \vee x_3 \vee x_4 \vee x_6 \vee x_7$). Do such c allow for "small" decision tree representations? (7)
- (c) Explain with an appropriate example how we can avoid over-fitting and under-fitting in constructing decision trees. (15)
- (d) What is VC dimension? How can we utilize such dimension in measuring the complexity of a learning machine? (6)
2. (a) Consider the following set of training examples: (5)

Instance	Classification	a_1	a_2
1	+	T	T
2	+	T	T
3	–	T	F
4	+	F	F
5	–	F	T
6	–	F	T

- (i) what is the entropy of this collection of examples with respect to the target function classification?
 - (ii) What is the information gain of a_2 relative to these training examples?
 - (b) Show that the minimum number of training examples m required for any consistent learner to successfully learn any target concept in H for any desired values of δ and ϵ is (15)
- $$m \geq \frac{1}{\epsilon} \left(\ln|H| + \ln\left(\frac{1}{\delta}\right) \right)$$
- (c) With the help of pseudo code describe FOIL algorithm for learning set of first-order rules. (15)

Contd P/2

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3. (a) Draw a flowchart showing different components of a learning machine and briefly describe each of them. (18)
- (b) What are the types of problems that can be solved with neural networks? What are the advantages and inconveniences of such networks? (11)
- (c) Is PAC learning a learning method or some notion describing the performance of a learning method? Give your justifications. (6)
4. (a) Discuss the principal properties of genetic algorithms (GAs). When should GAs be used and why? (10)
- (b) Explain the roles of encoding, crossover and mutation in GAs. (10)
- (c) You are hired by a match making web site to find matches between two groups of N people each (men and women). Each person has ordered its preferences from 1 to k in the other group. You should define a genetic algorithm that finds matches such that everyone gets the topmost possible choice. Describe the chromosome encoding, the fitness function and mutation operator for such a GA. (15)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Define consistent hypothesis and version space. State and prove the version space representation theorem. (12)
- (b) The manager of a cricket club evaluates a batsman by the following attributes for recruitment: (15)
- Reflex (possible values: Slow, Quick)
 - Range of Shots (possible values: Low, Medium, High)
 - Foot Work (possible values: Bad, Average, Good)
 - Running (possible values: Slow, Quick)
- Use CANDIDATE-ELIMINATION algorithm to learn the version space of the target concept based on the following training examples:
- (i) <Quick, Medium, Good, Slow>, Recruit = Yes
 - (ii) <Quick, High, Good, Slow>, Recruit = Yes
 - (iii) <Slow, Medium, Bad, Quick>, Recruit = No
 - (iv) <Quick, Medium, Good, Quick>, Recruit = Yes
- Show each step of your computation.
- (c) What are the advantages and disadvantages of k-nearest neighbor algorithm? (8)

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6. (a) Explain the various components of the Q learning algorithm with an illustrative example. (13)
- (b) Prove convergence of Q learning for deterministic Markov decision processes. How can you modify the training rule to ensure convergence of Q learning in a nondeterministic environment? (12)
- (c) How does reinforcement learning problem differ from other function approximation tasks? (10)
7. (a) Assume two different algorithms M1 and M2 were evaluated using cross-validation technique, in which the total number of examples in the data set is 500 and k is 10. The difference of sample error in the two algorithms were found and are shown in the following table: (18)

Iteration Number	No. of Samples Misclassified by M1	No. of Samples Misclassified by M2
1	7	8
2	6	4
3	7	10
4	9	9
5	8	9
6	9	6
7	10	3
8	8	11
9	7	9
10	6	5

Now compute the value of 95% confidence interval for the true difference of true error between the two algorithms. Assume that, the value of the t-statistic is 2.23.

- (b) Prove that $\sigma_{\text{error}_s(h)} = \sqrt{\frac{\text{error}_s(h)(1 - \text{error}_s(h))}{n}}$; where the symbols have their usual meanings. (12)

- (c) Write down three different criteria for the error function, E for Locally Weighted Linear Regression. Show the gradient weight update equation for each of these three criteria. (5)

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8. (a) How is instance based learning different from other learning methods? (7)
- (b) Does CANDIDATE-ELIMINATION algorithm always converge to the correct hypothesis? Explain your answer. (6)
- (c) Define lazy and eager learning. What are the advantages of lazy learning methods over eager methods? (8)
- (d) What difficulties arise when we try to estimate the future accuracy of a hypothesis given only a limited set of data? (7)
- (e) What are the limitations of the Find-S algorithm? How does Candidate-Elimination algorithm address these issues? (7)
-